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Python cmath Module

Python has a built-in module that you can use for mathematical tasks for complex numbers.

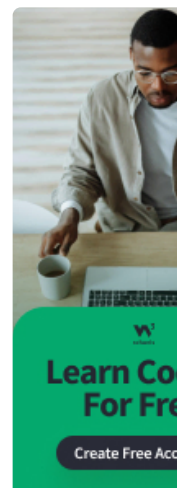
The methods in this module accepts `int`, `float`, and `complex` numbers. It even accepts Python objects that has a `__complex__()` or `__float__()` method.

The methods in this module almost always return a complex number. If the return value can be expressed as a real number, the return value has an imaginary part of 0.

The `cmath` module has a set of methods and constants.

cmath Methods

Method	Description
<code>cmath.acos(x)</code>	Returns the arc cosine value of x
<code>cmath.acosh(x)</code>	Returns the hyperbolic arc cosine of x
<code>cmath.asin(x)</code>	Returns the arc sine of x
<code>cmath.asinh(x)</code>	Returns the hyperbolic arc sine of x
<code>cmath.atan(x)</code>	Returns the arc tangent value of x
<code>cmath.atanh(x)</code>	Returns the hyperbolic arctangent value of x
<code>cmath.cos(x)</code>	Returns the cosine of x
<code>cmath.cosh(x)</code>	Returns the hyperbolic cosine of x

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<u>cmath.exp(x)</u>	Returns the value of E^x , where E is Euler's number (approximately 2.718281...), and x is the number passed to it
<u>cmath.isclose()</u>	Checks whether two values are close, or not
<u>cmath.isfinite(x)</u>	Checks whether x is a finite number
<u>cmath.isinf(x)</u>	Check whether x is a positive or negative infinity
<u>cmath.isnan(x)</u>	Checks whether x is NaN (not a number)
<u>cmath.log(x[, base])</u>	Returns the logarithm of x to the base base
<u>cmath.log10(x)</u>	Returns the base-10 logarithm of x
<u>cmath.phase()</u>	Return the phase of a complex number
<u>cmath.polar()</u>	Convert a complex number to polar coordinates
<u>cmath.rect()</u>	Convert polar coordinates to rectangular form
<u>cmath.sin(x)</u>	Returns the sine of x
<u>cmath.sinh(x)</u>	Returns the hyperbolic sine of x
<u>cmath.sqrt(x)</u>	Returns the square root of x
<u>cmath.tan(x)</u>	Returns the tangent of x
<u>cmath.tanh(x)</u>	Returns the hyperbolic tangent of x

cMath Constants

Constant	Description
<u>cmath.e</u>	Returns Euler's number (2.7182...)
<u>cmath.inf</u>	Returns a floating-point positive infinity value
<u>cmath.infj</u>	Returns a complex infinity value
<u>cmath.nan</u>	Returns floating-point NaN (Not a Number) value
<u>cmath.nanj</u>	Returns complex NaN (Not a Number) value
<u>cmath.pi</u>	Returns PI (3.1415...)
<u>cmath.tau</u>	Returns tau (6.2831...)